

K-Prototypes Clustering Project

AUTHOR

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Objective

Mental health research is a critical area of study, particularly in young people, however research at the population level is difficult. Mental health is not a single dimension, and there are most likely many routes to the same mental health problems, and the same factors likely lead to diverging issues. Traditional linear modelling, therefore, is limited in that it is focused on single outcomes. This is especially limiting in terms of policy, as knowledge of conditions can only be built on a condition-by-condition basis.

Partitioning-based clustering methods may offer useful exploratory insights in the case of mental health research in identifying groups for further study. K-means and k-modes are the most commonly used algorithms of this family, however only work when using only one of continuous (k-means) or categorical (k-modes) data. K-prototypes can work with both types of data, and therefore is the most applicable to psychological data, which, more often than not, involves both continuous and categorical data.

This project used a k-prototypes algorithm, aiming to identify clusters in an [online dataset](#) containing data on the mental health of young adults.

Methods

Python was used to perform a k-prototypes algorithm on the dataset linked above. Variables included were: age, gender, hours of sleep, stress level, physical activity, hours worked per day, mental health diagnosis, social interaction frequency, overall wellbeing score, diet quality, daily screen time, substance use, physical health conditions, and whether the participant had close friends. Analysis was performed using between 2 and 20 clusters in order to identify the optimal number of clusters.

Results

There was no clear optimal number of clusters, suggesting that the data were not in fact arranged into clear clusters. A reasonable person could choose anywhere between 4 and 7 clusters. This report chose 6, however it should be borne in mind that this may not necessarily be accurate. [Table 1](#) shows the mean for continuous variables and mode for categorical variables in each cluster. Because the sample was generally biased toward no mental or physical health diagnosis, drinking alcohol, and having close friends, the proportions for each of these variables are shown in [Table 2](#). There are no very clear differentiators between clusters, though it is worth noting that there were 2 clusters with higher rates of both depression (2 and 4) and anxiety (5 and 6). It may be useful to target young adults with these characteristics in future research on anxiety and depression.

Table 1

Table 1

A									
Cluster	Daily Screen Time (hrs)	Diet Quality	Wellbeing Score	Social Interaction	Daily Hours Worked	Physical Activity (mins per day)	Stress Level	Hours of Sleep	Age
1	7.0	2.7	4.8	0.9	7.5	30.3	4.7	7.7	23.2
2	7.8	1.8	6.5	0.8	7.7	28.9	3.1	6.2	28.7
3	6.8	2.9	2.9	1.0	8.6	27.7	6.7	5.8	28.2
4	7.6	1.3	2.7	0.9	8.1	30.7	6.8	6.7	23.0
5	5.4	1.5	4.9	0.9	8.9	30.4	4.7	7.7	29.5
6	8.1	1.8	2.7	0.8	7.2	24.8	6.8	7.3	31.6

B					
Cluster	Gender	MH Diagnosis	Substance Use	Has Close Friends	Physical Health Condition
1	Male	None	Alcohol	True	None
2	Female	None	Alcohol	True	None
3	Male	None	None	True	None
4	Female	None	Alcohol	True	None
5	Female	None	Alcohol	True	None
6	Male	None	Alcohol	True	None

Table 2

Table 2

A							
Cluster	Male	Has Close Friends	Depression	Anxiety	Drinks Alcohol	Drinks Alcohol and Smokes	Smokes
1	54.9%	92.2%	18.1%	13.0%	45.1%	11.4%	9.8%
2	42.0%	94.0%	24.0%	13.3%	39.3%	12.0%	9.3%
3	63.8%	91.3%	14.5%	13.0%	30.4%	12.3%	8.0%

A							
Cluster	Male	Has Close Friends	Depression	Anxiety	Drinks Alcohol	Drinks Alcohol and Smokes	Smokes
4	42.9%	89.5%	26.7%	14.1%	42.4%	12.0%	8.9%
5	42.4%	90.7%	19.2%	18.6%	40.7%	8.7%	14.5%
6	57.1%	93.6%	19.9%	19.2%	45.5%	9.6%	9.6%

B				
Cluster	Diabetes	Hypertension	Obesity	Other Chronic Illness
1	1.6%	7.8%	30.1%	6.7%
2	4.7%	7.3%	32.7%	6.7%
3	2.9%	10.1%	23.9%	10.1%
4	2.1%	8.9%	27.7%	5.2%
5	1.7%	12.2%	33.7%	4.7%
6	4.5%	7.1%	27.6%	10.3%